

COMPUTER HARDWARE AND SOFTWARE



WHAT IS COMPUTER HARDWARE?

- Is an umbrella term that refers to the physical components of a computer.
- Encompasses all the physical, tangible components required for a computer to operate.
- These parts are visible and touchable, divided into two main categories based on location:
 - **Internal Hardware:** Essential components located inside the device that enable it to run, such as the central processing unit (CPU), memory, and video card.
 - **External Hardware:** Peripherals located outside or around the device that enhance usability and functionality, such as a keyboard or computer case.

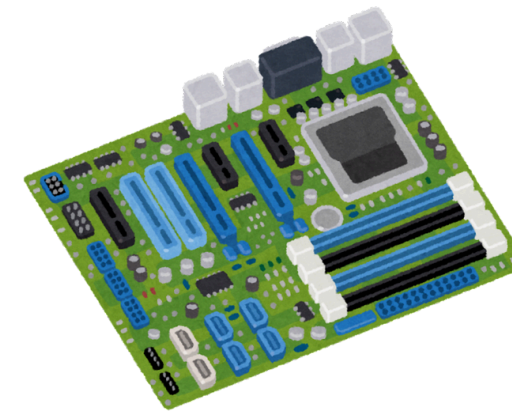
FOUR PRIMARY CATEGORIES OF COMPUTER HARDWARE

The hardware components of a computer or personal computer is categorized into 4 primary categories:

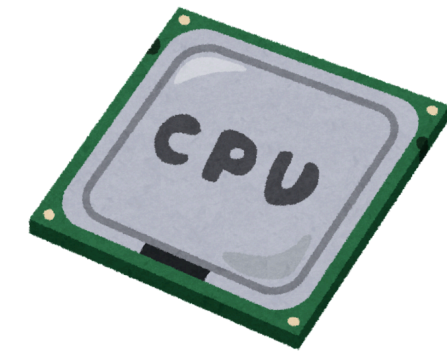
- **System Unit**
- **Display Device**
- **Input Devices**
- **External Devices**

SYSTEM UNIT

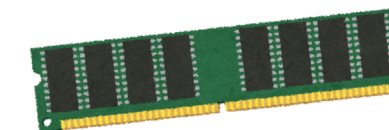
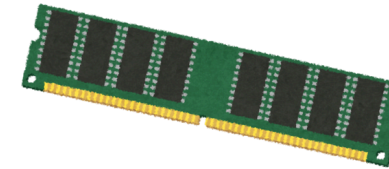
- A System Unit is the main component of a personal computer, which houses the other devices necessary for the computer to function.
- It is comprised of a chassis and the internal components of a personal computer such as the system board (mother board), the microprocessor, memory modules, disk drives, adapter cards, the power supply, a fan or other cooling device and ports for connecting external components such as monitors, keyboards, mice, and other devices.



MOTHERBOARD



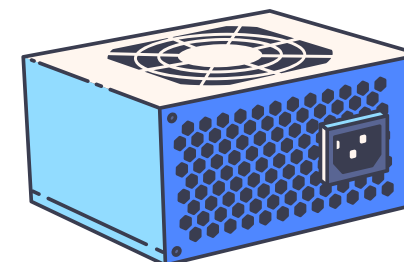
CPU



RAM



STORAGE DEVICE



POWER SUPPLY

DISPLAY DEVICE

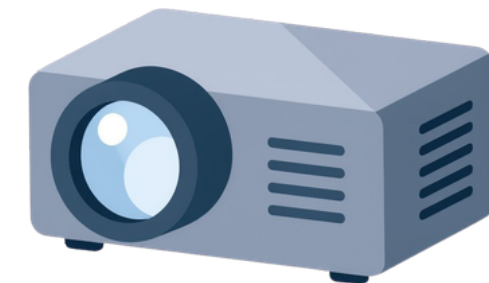
- A display device is a personal computer component and is an output device that enables user to view the text and graphical data associated with a computer program.
- Display devices commonly connect to the system unit via a cable, and they have controls to adjust the settings for the device.
- They vary in size and shape, as well as the technology used.



MONITOR



LCD DISPLAY



PROJECTOR

INPUT DEVICES

- An input device is a personal computer component that enables users to enter data or instructions into a computer.
- The most common input devices are keyboards and computer mice. Input devices can connect to the system via a cable or a wireless connection.



KEYBOARD



MOUSE



MICROPHONE



SCANNER



WEBCAM



TOUCHSCREEN

EXTERNAL DEVICES

- Any peripherals devices that are not housed inside the system unit are inherently external devices.
- A personal computer's functionality can be enhanced by connecting different types of external devices to the system unit, often called peripherals devices.
- These devices typically provide alternative input or output methods or additional data storage.
- External devices are connected to the system unit via a cable or wireless connection. Some of them have their own power source and some draw power from the system.



PRINTER



SPEAKER



USB FLASH DRIVE



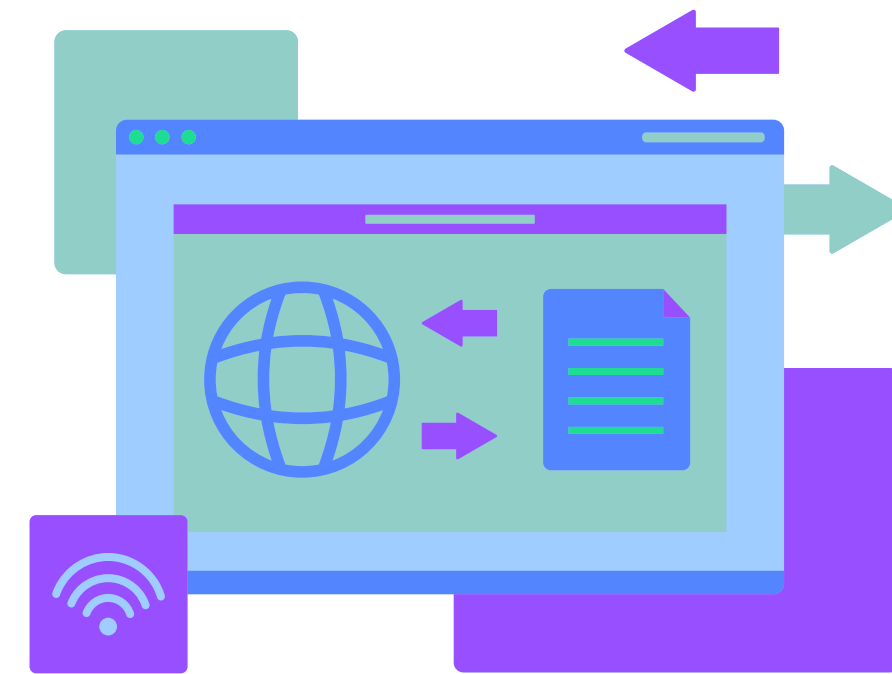
EXTERNAL HARD DRIVE



WEBCAM

WHAT IS SOFTWARE?

- Is a collection of instructions, data, or programs that tell a computer how to work and execute specific tasks.
- Unlike physical computer hardware (the parts you can touch), software is the intangible set of code that makes hardware useful.
- **Example:**
 - Operating System
 - Web Browser
 - Microsoft Word
 - Games
 - Mobile Apps



TYPES OF SOFTWARE

- System Software
 - Operating System
 - Language Processor
 - Device Driver
- Application Software
 - General Purpose Software
 - Customize Software
 - Utility Software

SYSTEM SOFTWARE

- Is a software that directly operates the computer hardware and provides the basic functionality to the users as well as to the other software to operate smoothly.
- System Software basically controls computer's internal functioning and also controls hardware devices such as monitors, printer and storage devices, etc.

TYPES OF SYSTEM SOFTWARE

Operating System

- It is the main program of a computer system.
- When the computer system ON it is the first software that loads into the computer's memory
- It manages all the resources such as computer memory, CPU, printer, hard disk, etc., and provides an interface to the user, which helps the user to interact with the computer system. It also provides various services to other computer software.

TYPES OF SYSTEM SOFTWARE

Language Processor

- Translate programs written in high-level languages into machine code that a computer understands and assemblers translate programs written in low-level or assembly language into machine code.
- In the compilation process, there are several stages.

TYPES OF SYSTEM SOFTWARE

Device Driver

- Is a program or software that controls a device and helps that device to perform its functions.
- Components like printers, mice, and modems require these drivers to communicate with the operating system, meaning users typically need to install them so the computer can successfully recognize and manage the device.

FEATURES OF SYSTEM SOFTWARE

- System Software is closer to the computer system
- System Software is written in a low-level language in general
- System Software is difficult to design and understand
- System Software is fast in speed (working speed)
- System Software is less interactive for the users in comparison to application software

APPLICATION SOFTWARE

- Software that performs special functions or provides functions that are much more than the basic operation of the computer is known as **application software**.
- Is designed to perform a specific task for end-users. It is a product or a program that is designed only to fulfill end-users' requirements.
- It includes word processors, spreadsheets, database management, inventory, payroll programs, etc.

TYPES OF APPLICATION SOFTWARE

General Purpose Software

- This type of application software is used for a variety of tasks and it is not limited to performing a specific task only.
- For example, MS-Word, MS-Excel, PowerPoint, etc.

TYPES OF APPLICATION SOFTWARE

Customized Software

- This type of application software is used or designed to perform specific tasks or functions or designed for specific organizations.
- For example, railway reservation system, airline reservation system, invoice management system, etc.

TYPES OF APPLICATION SOFTWARE

Utility Software

- This type of application software is used to support the computer infrastructure. It is designed to analyze, configure, optimize and maintain the system, and take care of its requirements as well.
- For example, antivirus, disk fragmenter, memory tester, disk repair, disk cleaners, registry cleaners, disk space analyzer, etc.

DIFFERENCE OF SYSTEM SOFTWARE AND APPLICATION SOFTWARE

System Software	Application Software
It is designed to manage the resources of the computer system, like memory and process manangement, et.c	It is designedd to fulfill the requirements of the user for performing specific tasks.
Written in a low-level language	Written in a high-level language
Less interactive for the users.	More interactive for the users
System software plays vital role for the effective functioning of a system	Applicatopn software is not to important for the fuunctioning of the system, as it is specific
It is independent of the application software to run	It needs system software to run

DIFFERENCE OF HARDWARE AND SOFTWARE

Software	Hardware
Software consists of intangible sets of instructions, data, programs, and code that tell the hardware what to do. You cannot physically touch software.	Hardware refers to the physical, tangible parts of a computer system that you can actually touch and see
Software cannot run without underlying hardware to host and execute it.	Hardware cannot perform any useful work without software to instruct it.
Less interactive for the users.	Hardware is manufactured using physical electronic components and materials, and repairing or upgrading it often requires replacing physical parts.
Software is developed through computer programming languages and can be easily updated, modified, or reinstalled digitally via code.	Application software is not too important for the functioning of the system, as it is specific



THANK YOU